Inflation Drivers in Firms' Words: Bridging Micro Narratives and Macro Dynamics

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Disclaimer: The views expressed are those of the authors and do not necessarily reflect those of the Bank of Canada.

Firm narratives reveal inflation drivers

Question

What can firms' own explanations tell us about the drivers of inflation?

- Standard macro data show prices moved.
- Firm narratives identify **why** firms changed prices.
- We use LLMs to turn these narratives into structured macro data.



Firm narratives separate salience from inflation contribution.

- Demand is salient and frequent in reports, but positive and negative attributions are nearly symmetric. It plays a limited role in COVID inflation.
- Non-labor input costs dominate the main inflation surge.
- Wage pressure matters later and helps explain persistence.

What do attribution narratives offer?

A simple illustration (details later)

$$\underbrace{\Pi_t}_{\text{Aggregated inflation}} = \theta_0 + \sum_{m=1}^{k} \underbrace{\theta_m}_{\text{Factor m loading}} \underbrace{1(|X_{m,t}| > k_m)}_{\text{Conditional trigger}} X_{m,t} + \epsilon_t$$

- Correct conditioning of $1(|X_{m,t}| > k_m)$ is necessary to identify θ_m
- Attribution ("why") narratives provide a measure of $1(|X_{m,t}| > k_m)$

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Clarification: we see m as factors

- reflecting firms' own localized decision-relevant conditions
- $\neq$ macro structural shocks
- $\neq$ general equilibrium conditions

Related literature

- **Text and narratives as macro data**
  Gentzkow et al. (2019); Larsen and Thorsrud (2019); Bybee et al. (2021); Andre et al. (2022)
  ↪ we focus on micro narratives

- **Firms' pricing reasons and pass-through**
  Blinder et al. (1998); Dogra et al. (2023); Gagliardone et al. (2025); Hoeck and Renkin (2025)
  ↪ we focus on extensive-margin adjustment triggered by *multiple* factors

- **Post-COVID inflation and state-dependent repricing**
  Bernanke and Blanchard (2025); Giannone and Primiceri (2024); Lorenzoni and Werning (2023); Cavallo et al. (2024); Montag and Villar (2025)
  ↪ we pursue a bottom-up (not top-down) view by "listening to firms"

Firms explain price changes in real time

- 3300+ regional Beige Book reports, 1990–2024.
- Real-time descriptions of firm and market-contact conditions.
- Dense, stable, economically focused language.
- We focus on firm pricing conditions and related factors.
- The Beige Book often reports not only what changed, but also why firms say it changed.

Attributions are the key measurement object

"[Parcel companies have increased shipping prices] {price} due to [unprecedented demand from online shopping overall and holiday demand specifically] {demand}."

January 2021 Beige Book, St. Louis Fed

LLM extraction

- economic concept: price, demand, wage, input cost...
- direction: $+, ++, +++, -, --, ---, =$
- time: realized or expected
- sector: manufacturing, retail, services...

Object used for macro

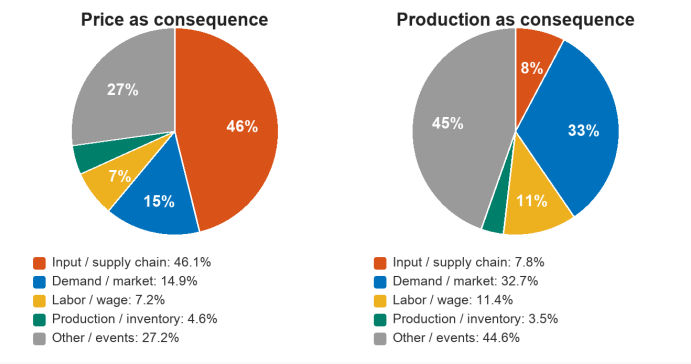
$$\textit{demand} \uparrow \quad \Rightarrow \quad \textit{price} \uparrow$$

- phrase: a factor is mentioned,
- attribution: the factor is linked to price adjustment.
- **Key distinction:** phrase salience $\neq$ price attribution.

Extracted attributions are structured macro data

Objects extracted

- Classified variables: 323,676.
- Total attributions: 10,403.
- Human validation: phrases 84.6%, attributions 91.87%.
- Attributions by outcomes: price 47.0%, production 20.6% wage 20.5%, profit margin 11.9%.



validation | word clouds | attribution summary | broader network

Directional attributions match macro intuition

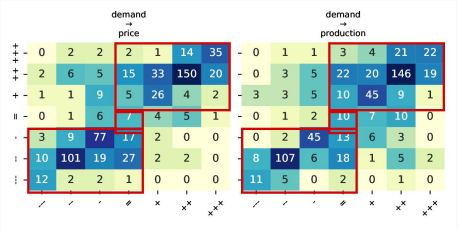
Directional Attributions

- Each panel fixes:
  - Vertical axis: **cause**;
  - Horizontal axis: **consequence**;
  - **Price** (left) vs. **production** (right) consequences.
- Dark cells are more frequent directional links; red boxes mark theory-consistent regions.
- Input costs: $c \uparrow \Rightarrow p \uparrow, q \downarrow$. 91.8% consistent.
- Demand: $d \uparrow \Rightarrow p \uparrow, q \uparrow$. 93.2% consistent.
- Wage attributions are also highly consistent: 92.7%.

Input-cost attributions

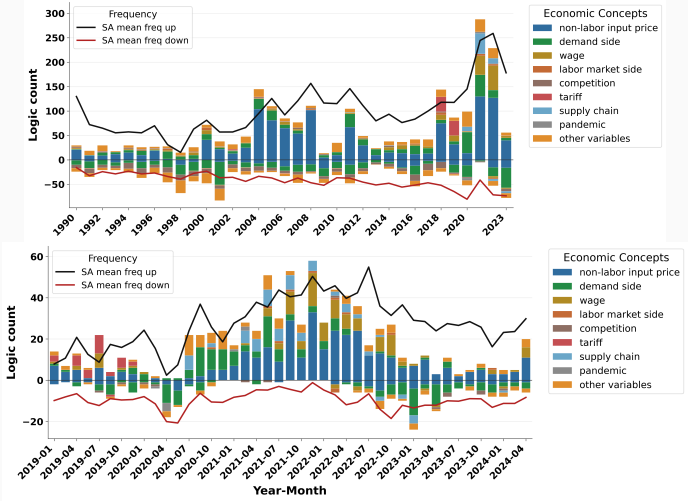


Demand attributions



Price reasoning aligns with repricing frequency

- Price attributions record reasoning for repricing, not price-change size.
- Counts move with micro repricing frequency.
- Upward reasoning rises in the post-COVID repricing surge.
- Salient factors change over time: demand → input costs → wages.
- Freq of price attributions is not an inflation contribution.



Bars: price attributions; lines: repricing freq (Montag and Villar (2025))

From attribution counts to structural contributions

Issue with reduced-form attributions	Structural object that resolves it
Pricing behavior. Reports name pressures, but firms adjust only when pressure is large enough, with firm heterogeneity.	**Menu-cost decision:** aggregate $z_{m,t}$ and firm-level $v_{i,m,t}$ determine threshold crossing; reporting friction links adjustments to observed attributions.
Aggregation. Attribution counts are frequencies, not inflation contributions.	**Inflation equation:** $M^{\pm}_{m,t}$ combines crossing mass and active gap size; ψ_m puts factor pressure in inflation units.
Latent pressure. Text, macro series, and inflation are noisy signals of the same underlying factor.	**State-space likelihood:** filter $z_{m,t}$ and estimate thresholds, loadings, and reporting noise jointly.

- **Therefore:** use narrative attributions structurally, not as reduced-form contribution shares.

Three factor pressures move firms' desired prices

Flexible desired price

$$p^0_{i,t}: \ \max_{P,L,M} \ P_{i,t}Y_{i,t} - W_{i,t}L_{i,t} - Q_{i,t}M_{i,t}$$

primitive	pricing channel	price term
non-homothetic demand: $P_{i,t} = \tilde{\lambda}_t \omega_{i,t} u'(Y_{i,t})$	$\tilde{\lambda}_t, \omega_{i,t}$ shift demand and market power	$\beta_\lambda \hat{\lambda}_t$, $\beta_\omega \hat{\omega}_{i,t}$
non-labor input cost $Q_{i,t}$	shifts marginal cost	$\beta_q q_t$, $\beta_q e^q_{i,t}$
wage $W_{i,t}$	shifts marginal cost	$\beta_w w_t$, $\beta_w e^w_{i,t}$

$$p^0_{i,t} = \underbrace{\beta_\lambda \hat{\lambda}_t + \beta_q q_t + \beta_w w_t}_{\Phi_t\text{: aggregate pressure}} + \underbrace{\beta_\omega \hat{\omega}_{i,t} + \beta_q e^q_{i,t} + \beta_w e^w_{i,t}}_{p^*_{i,t}\text{: firm-level pressure}}.$$

Menu-cost decision

$$\min_{\{p_{i,t}\}} \mathbb{E}_0 \sum_t \left[\tfrac{1}{2}(p_{i,t} - p^0_{i,t})^2 + \xi 1(p_{i,t} \neq p_{i,t-1}) \right].$$

$$p_{i,t} = \begin{cases} p^0_{i,t}, & p_{i,t-1} - p^0_{i,t} \notin [g^L, g^U], \\ p_{i,t-1}, & p_{i,t-1} - p^0_{i,t} \in [g^L, g^U]. \end{cases}$$

Firms adjust only when the inherited price gap exits the inaction band.

micro primitives

Threshold crossings define inflation objects

State-space factor representation

- Factor-m pricing pressure: $s^L_{i,m,t} = z_{m,t} + v_{i,m,t}$.
- State law: $z_{m,t} = \rho_m z_{m,t-1} + \eta_{m,t}$; $v_{i,m,t}$ is firm-level.

m	factor	aggregate state $z_{m,t}$	pass-through
d	demand	$\hat{\lambda}_t$	β_λ
c	non-labor input	q_t	β_q
w	wage	w_t	β_w

- Trigger rule: $s^L_{i,m,t} > \kappa_{m,+}$ or $s^L_{i,m,t} < -\kappa_{m,-}$.
- Cutoff link: $\kappa_{m,+} = -g^L/\beta_m, \quad \kappa_{m,-} = g^U/\beta_m$. Larger β_m lowers the cutoffs.

From thresholds to inflation units

$$M^+_{m,t}(z) = \mathbb{E}\left[s^L_{i,m,t} 1\{s^L_{i,m,t} \geq \kappa_{m,+}\} \mid z_{m,t}\right],$$
$$M^-_{m,t}(z) = \mathbb{E}\left[s^L_{i,m,t} 1\{s^L_{i,m,t} \leq -\kappa_{m,-}\} \mid z_{m,t}\right].$$

$$\Pi^{nar}_t = \theta_0 + \sum_m \psi_m \left[M^+_{m,t}(z_{m,t}) + M^-_{m,t}(z_{m,t})\right].$$

- κ: extensive margin.
- $M^{\pm}$: crossing mass and active gap size.
- ψ: intensive loading into inflation units.

cutoff details | moment details

Reports are noisy signals of pricing pressure

Aggregate factor

$$z_{m,t} = \rho_m z_{m,t-1} + \eta_{m,t}$$

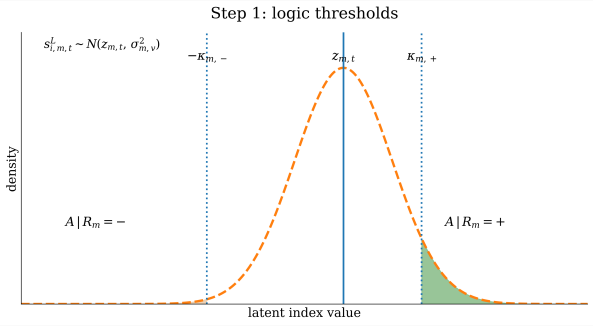
Attribution reports: conditional frequency

$$s^L_{i,m,t} = z_{m,t} + v_{i,m,t}, \qquad v_{i,m,t} \sim \mathcal{N}(0, \sigma^2_{v,m})$$

$$L^{\pm}_{m,t} \mid N^{\pm}_{m,t}, z_{m,t} \sim \text{Binomial}\big(N^{\pm}_{m,t}, q^{\pm}_m(z_{m,t})\big)$$

$$q^+_m(z) = \Pr(s^L_{i,m,t} > \kappa_{m,+} \mid z), \quad q^-_m(z) = \Pr(s^L_{i,m,t} < -\kappa_{m,-} \mid z)$$

- $L^{\pm}/N^{\pm}$: conditional attribution frequency.
- $q^{\pm}_m(z)$: threshold-crossing probability.
- identifies $\kappa_{m,\pm}/\sigma_{v,m}$ (extensive margin).



Attribution reports identify conditional threshold probabilities

Reports are noisy signals of pricing pressure

Aggregate factor

$$z_{m,t} = \rho_m z_{m,t-1} + \eta_{m,t}$$

Phrase reports: salience and inflation

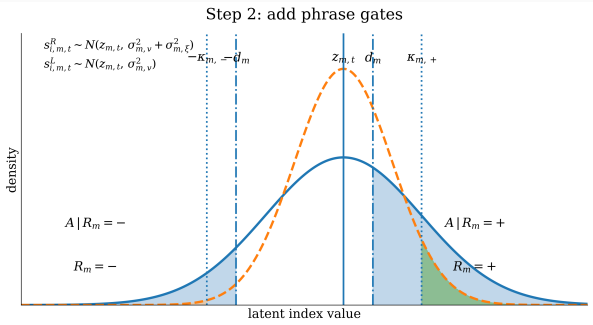
$$s^R_{i,m,t} = s^L_{i,m,t} + \xi_{i,m,t}, \qquad \xi_{i,m,t} \sim \mathcal{N}(0, \sigma^2_{m,\xi})$$

$$(N^+_{m,t}, N^0_{m,t}, N^-_{m,t}) \mid z_{m,t} \sim \text{Multinomial}\big(p^+_m(z), p^0_m(z), p^-_m(z)\big)$$

$$\Pi_t = \theta_0 + \textstyle\sum_m \psi_m \left[M^+_{m,t}(z_{m,t}) + M^-_{m,t}(z_{m,t})\right] + \varepsilon^\Pi_t$$

- phrases define $N^\pm$ and $\sigma_{m,\xi}$;
- inflation identifies the loading ψ_m.
- all measurements help infer $z_{m,t}$.

MLE/GEM: jointly estimates parameters and filters factor states.



Phrase reports define salience; inflation sets ψ_m

likelihood details

Estimates pin down triggers, loadings, and states

Read from estimates

- Input costs: easiest trigger and cleanest signal.
- Wages: largest loading and most persistent state.
- Demand: smaller loading and noisier signal.

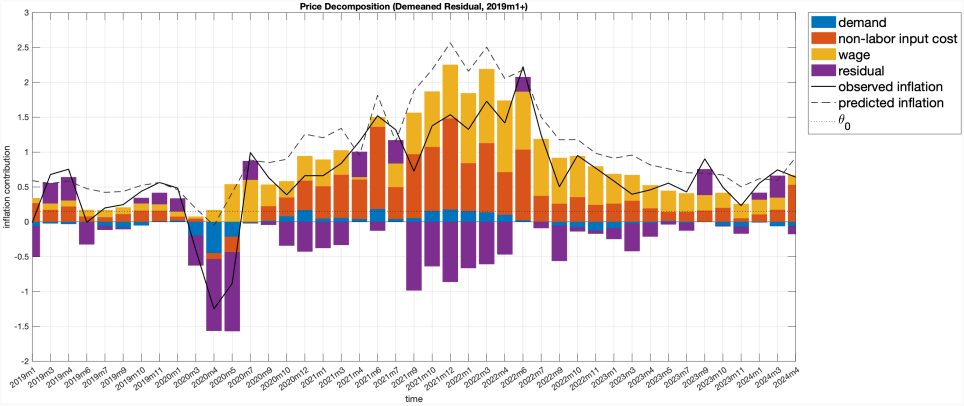
parameter table | trigger frequencies

filtered state

object	parameter	demand	input cost	wage
extensive margin	κ^+/σ_v	2.53	2.24	2.76
intensive margin	ψ	0.42	0.70	0.81
state persistence	ρ	0.61	0.83	0.95
reporting noise	σ_ξ	5.07	0.02	18.13

- Lower κ: more firms cross the repricing threshold.
- Larger ψ: stronger pass-through to inflation.
- Larger ρ: the state remains elevated longer.

Input costs drive the surge; wages matter later



Model-based decomposition, post-2019 | shares: Demand 13.86%, Input cost 41.96%, Wage 25.83%, Residual 18.36% (81.64% of post-2019 variation). full sample

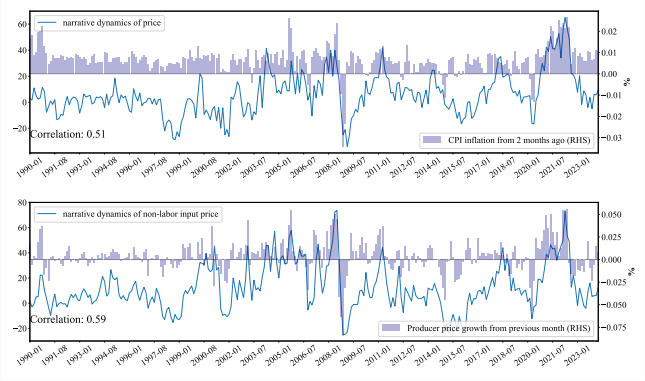
Conclusion

- Firm text can identify **reasons for price adjustment**, not just sentiment.
- Attributions are informative because pricing is lumpy: they reveal **repricing triggers**.
- A structural model turns trigger attributions into inflation contributions.
- Post-COVID inflation:
  - demand is salient but limited,
  - non-labor input costs dominate the surge,
  - wage pressure matters for persistence.

Validation: narrative dynamics versus macro series

Key results

- Narrative price dynamics co-move with CPI.
- Narrative input-price dynamics co-move with PPI.
- Text measures capture direction/frequency rather than exact magnitude.



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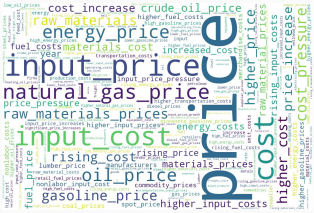
Word clouds

Concept validity

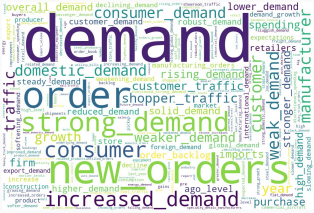
- Price phrases: inflation, retail price, consumer price.
- Input cost: raw materials, energy, commodities.
- Demand: orders, consumer demand.
- Wage: labor cost, wage pressure.



(a) Price



(b) Input cost



(c) Demand



(d) Wage

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Directional attribution: factor ⇒ consequence

Economic content

- Input costs mainly show up in price narratives.
- Demand mainly shows up in output/activity narratives.
- Labor conditions mainly show up in wage narratives.

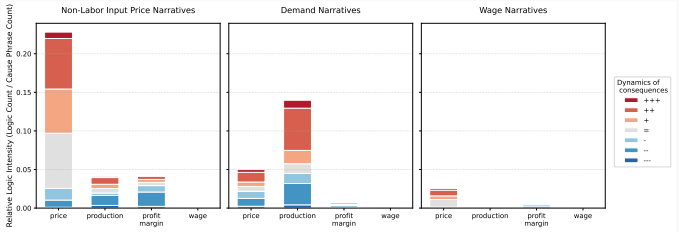


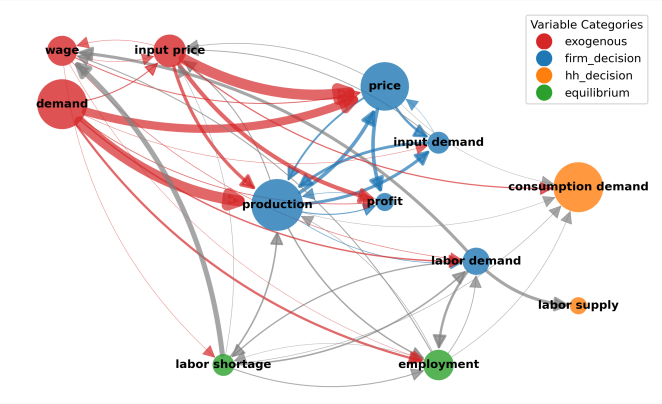
Figure: Consequences by cause

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Broader attributions reveal the full narrative network

Broader extraction

- Unrestricted attribution extraction: 35,563 attribution chains.
- Attribution types: firm P.E. 76%, G.E. 16%, household 8%.
- Most attributions describe firm's partial equilibrium decision-making.



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Micro primitives generate factor-specific price pressure

Demand and markup

$$P_{i,t} = \tilde{\lambda}_t \, \omega_{i,t} u'(Y_{i,t}), \qquad Y_{i,t} = (u')^{-1}\left(\frac{P_{i,t}}{\tilde{\lambda}_t \omega_{i,t}}\right).$$

$$\mu(Y) = \frac{\varepsilon(Y)}{\varepsilon(Y) - 1}, \qquad \varepsilon(Y) = -\frac{u'(Y)}{u''(Y)Y}.$$

- $\tilde{\lambda}_t, \omega_{i,t} \uparrow \Rightarrow Y_{i,t} \uparrow$.
- Non-homotheticity makes $\mu(Y_{i,t})$ move with demand.

Marginal cost and linearization

$$MC_{i,t} = \frac{1}{A_{i,t}} \left(\frac{W_{i,t}}{\alpha}\right)^{\alpha} \left(\frac{Q_{i,t}}{1-\alpha}\right)^{1-\alpha}$$

$$P^0_{i,t} = \mu(Y_{i,t}) MC_{i,t} \Rightarrow p^0_{i,t} = \beta_\lambda \hat{\lambda}_t + \beta_q q_t + \beta_w w_t + p^*_{i,t}.$$

- Non-labor input prices and wages shift marginal cost.
- $p^*_{i,t}$ collects firm-specific demand and cost terms.
- β_m is structural pass-through into desired prices.

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Menu-cost cutoff details

Factor-alone trigger

$$g^L < 0 < g^U.$$

$$p^0_{i,t} = \beta_m z_{m,t} + \beta_m v_{i,m,t} + \text{other factors}.$$

$$s^L_{i,m,t} = z_{m,t} + v_{i,m,t}.$$

$$\beta_m s^L_{i,m,t} > -g^L \quad \Rightarrow \quad s^L_{i,m,t} > \kappa_{m,+}.$$

$$\beta_m s^L_{i,m,t} < -g^U \quad \Rightarrow \quad s^L_{i,m,t} < -\kappa_{m,-}.$$

$$\kappa_{m,+} = -g^L/\beta_m, \qquad \kappa_{m,-} = g^U/\beta_m.$$

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Factor-specific mapping

factor	$z_{m,t}$	$v_{i,m,t}$	$(\kappa_{m,+}, \kappa_{m,-})$
demand	$\hat{\lambda}_t$	$\frac{\beta_\omega \hat{\omega}_{i,t} - p_{i,t-1}}{\beta_\lambda}$	$\left(-g^L/\beta_\lambda,\ g^U/\beta_\lambda\right)$
non-labor input cost	q_t	$\frac{\beta_q e^q_{i,t} - p_{i,t-1}}{\beta_q}$	$\left(-g^L/\beta_q,\ g^U/\beta_q\right)$
wage	w_t	$\frac{\beta_w e^w_{i,t} - p_{i,t-1}}{\beta_w}$	$\left(-g^L/\beta_w,\ g^U/\beta_w\right)$

Active threshold crossings and inflation moments

Active adjustment

$$A^{+}_{i,m,t} = 1\{s^{L}_{i,m,t} \geq \kappa_{m,+}\},$$
$$A^{-}_{i,m,t} = 1\{s^{L}_{i,m,t} \leq -\kappa_{m,-}\}.$$
$$\Delta p^{FA}_{i,m,t} = \beta_m s^{L}_{i,m,t} A^{+}_{i,m,t} + \beta_m s^{L}_{i,m,t} A^{-}_{i,m,t}.$$

- Counterfactual holds other forces at baseline.
- Active adjustment uses both crossing and gap size.

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Aggregate truncated moments

$$M^{+}_{m,t}(z) = \mathbb{E}\left[s^{L}_{i,m,t} 1\{s^{L}_{i,m,t} \geq \kappa_{m,+}\} \mid z_{m,t}\right]$$
$$M^{-}_{m,t}(z) = \mathbb{E}\left[s^{L}_{i,m,t} 1\{s^{L}_{i,m,t} \leq -\kappa_{m,-}\} \mid z_{m,t}\right]$$
$$\Pi^{nar}_{t} = \theta_0 + \sum_{m} \psi_m \left[M^{+}_{m,t}(z_{m,t}) + M^{-}_{m,t}(z_{m,t})\right].$$

- $M^{\pm}$: extensive margin with active gap size.
- ψ_m: inflation-unit loading.

Likelihood detail: reporting probabilities

Phrase gates

Suppress m, t subscripts.

$$s^R = z + v + \xi, \qquad \sigma_R^2 = \sigma_v^2 + \sigma_\xi^2.$$

$$p_R^+(z) = 1 - \Phi(b^+), \quad b^+ = \frac{d - z}{\sigma_R}.$$

$$p_R^-(z) = \Phi(b^-), \quad b^- = \frac{-d - z}{\sigma_R}.$$

$$(N_+, N_0, N_-) \mid z \sim \text{Multinomial}\Big(N; p_R^+, 1 - p_R^+ - p_R^-, p_R^-\Big).$$

Attribution count likelihood

$$M_{A,\pm} \mid N_\pm, z \sim \text{Binomial}\big(N_\pm, q_{A|R\pm}(z)\big).$$

Attribution within phrase branches

$$s^L = z + v, \qquad \rho = \frac{\sigma_v}{\sigma_R}.$$

$$a^+ = \frac{\kappa_+ - z}{\sigma_v}, \qquad a^- = \frac{-\kappa_- - z}{\sigma_v}.$$

$$q_{A|R+}(z) = \frac{1 - \Phi(a^+) - \Phi(b^+) + \Phi_2(a^+, b^+; \rho)}{1 - \Phi(b^+)}$$

$$q_{A|R-}(z) = \frac{\Phi_2(a^-, b^-; \rho)}{\Phi(b^-)}$$

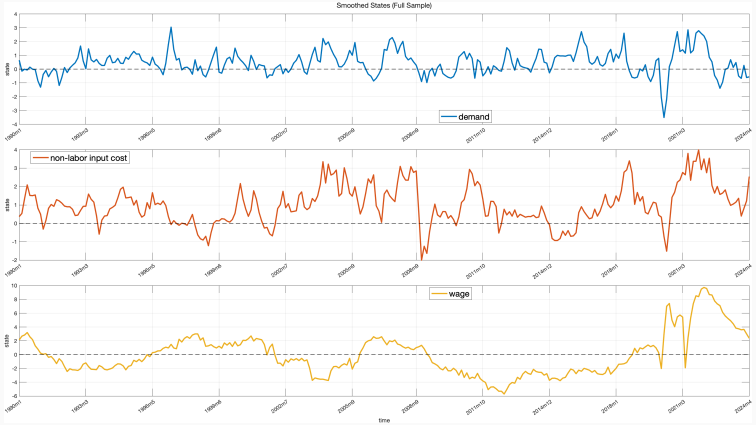
- Φ_2: standard bivariate normal CDF.
- Positive branch uses the joint upper-tail intersection.

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Filtered states summarize factor timing

State interpretation

- $z_{m,t}$ is the filtered factor state.
- It combines text signals and macro series.
- Persistence matters for inflation contribution.



Smoothed latent states from the estimated state-space model

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Estimated parameters

	Attribution/State block						Price block		Signal block	
factor	ρ_m	$\sigma_{v,m}$	$\sigma_{\xi,m}$	d_m	κ_m^+	κ_m^-	ψ_m	θ_0	R_P	c_m
demand	0.6069	3.4970	5.0695	0.9808	8.8480	7.7287	0.4200	0.1468	0.2869	0.1433
	(0.063)	(0.249)	(0.375)	(0.053)	(0.571)	(0.520)	(0.171)	(0.071)	(0.025)	(0.035)
non-labor input cost	0.8315	2.4874	0.0238	0.5501	5.5716	5.6167	0.7024	0.1468	0.2869	0.2189
	(0.032)	(0.164)	(0.241)	(0.039)	(0.362)	(0.373)	(0.073)	(0.071)	(0.025)	(0.023)
wage	0.9517	9.1218	18.1277	6.7867	25.1656	–	0.8114	0.1468	0.2869	0.3279
	(0.018)	(1.267)	(1.606)	(0.410)	(3.460)	–	(0.514)	(0.071)	(0.025)	(0.004)

- κ: extensive-margin trigger.
- ψ: intensive-margin loading.
- σ_ξ: phrase noise relative to attribution.

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Model-implied trigger frequencies

Key results

- Input-cost triggers are most frequent on the price margin.
- Demand triggers are more balanced across upward and downward directions.
- This helps explain why demand has a smaller net inflation contribution.

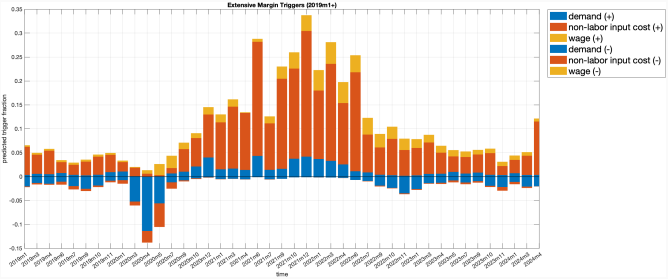


Figure: Model-implied trigger frequencies, post-2019

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Full-sample decomposition

Full sample shares

- Demand: 5.22%
- Non-labor input cost: 25.85%
- Wage: 12.17%
- Residual: 56.76%

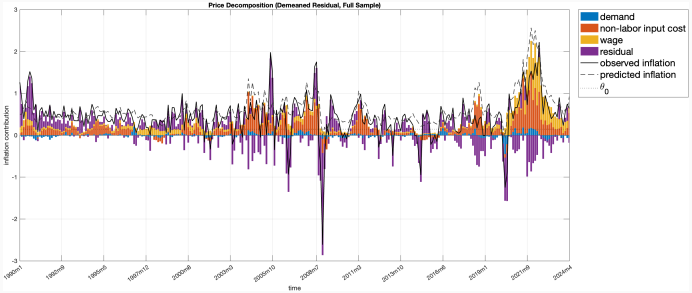


Figure: Model-based decomposition, full sample

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